# Kian Maleki, PhD — Data Scientist | Machine Learning | Statistical Modeling

+1 (737) 600 9803 | kian@drkianmaleki.com | drkianmaleki.com | linkedin.com/in/kian-maleki-phd | github.com/drkianmaleki
University of Iowa (2025) PhD in Physics (GPA: 4.03) | Creighton University (2017) MS in Physics (GPA: 4.00)

## Profile

- Data scientist with a physics PhD and ten years of research experience in statistical inference, mathematical modeling, and scientific computing.
- Builds end-to-end: open-source Python packages, automated testing and CI, containerized and reproducible pipelines from raw data to evaluated model.

## Technical Expertise

- **Machine Learning & Statistics:** Bayesian inference, Monte Carlo simulation, classifier evaluation, calibration, bootstrap confidence intervals, XGBoost, scikit-learn, PyTorch, TensorFlow.
- **Data & Software Engineering:** Python, SQL, Pandas, Polars, NumPy, Numba, FastAPI, Pydantic, Linux/Bash, HPC.
- **MLOps & Cloud:** Docker, Git, GitHub Actions, MLflow, DVC, Evidently, AWS SageMaker, S3, Lambda, CloudWatch, IAM.
- **Generative AI & NLP:** RAG, Transformers, document intelligence, Bedrock.

## Training & Certifications

**BlueDot Impact | Technical AI Safety Project** Aug 2026 – Sep 2026

- Completed the program research project: two studies of chain-of-thought in reasoning models, *Decided Mid-Thought* and *Reasoning Trace Experiments*.

**AI-Powered Document Intelligence Extern | Pfizer-Sponsored Externship** Apr 2026 – Jul 2026

*link:* https://github.com/drkianmaleki/PharmaDoc-AI

- Built PharmaDoc-AI, the externship capstone: a prototype automating document extraction, retrieval, and question answering for healthcare supply-chain use cases with native PDF text extraction, LLMs, and RAG architectures.

**TripleTen | AI & Machine Learning Program (36 weeks)** Nov 2025 – Apr 2026

- Completed 15+ end-to-end ML projects spanning supervised and unsupervised learning, deep learning, computer vision, NLP, model evaluation, SQL, and Git.

## Professional Experience

**Independent Machine Learning Engineer | Open-Source Research & Development** Aug 2025 – Present

- Design and ship end-to-end open-source machine-learning software as tested Python packages with pytest, CI via GitHub Actions, and reproducible experiment pipelines: a classifier-evaluation toolkit, a learning-curve benchmarking harness, a RAG document assistant, and a domain-specific product-name normalization engine.
- Completed a pre-registered causal study of chain-of-thought faithfulness in a reasoning model (*Decided Mid-Thought*), using prefix interventions and 500 resampled continuations to test when the final answer becomes determined.
- Engineered a reproducible benchmark evaluating 51 sequence-acceleration methods from 13 families across 18 convergence regimes for finite-horizon learning-curve prediction in gradient boosting, with automated failure detection and conservative extrapolation rules for operational use.
- Implemented ambiguity metrics, calibration analysis, bootstrap confidence intervals, and paired statistical tests to quantify when model comparisons and predictions can be trusted.

**Creighton University | Resident Assistant Professor of Physics** Aug 2025 – Jul 2026

- Taught undergraduate physics courses (40–180 students), delivering lectures on optics and electronics with a focus on applied problem-solving.
- Supervised and mentored 1–5 teaching assistants, improving instructional quality and consistency in grading.
- Developed interdisciplinary lab curriculum integrating AI tools and chatbots into scientific workflows.

**University of Iowa | Doctoral Researcher, Physics** Aug 2018 – Aug 2025

- Optimized large-scale numerical simulations, reducing computational runtime from days to under one hour through algorithmic improvements.
- Developed high-dimensional modeling frameworks for complex physical systems using statistical and numerical methods.
- Analyzed high-dimensional experimental data to extract signal from noise using statistical inference and optimization.

## Selected Projects

**Chain-of-Thought Studies: Reasoning Trace Experiments and Decided Mid-Thought** 2026

*Tech:* Python, LLM APIs (OpenRouter/DeepInfra/Claude), Resampling, LLM-as-Judge, Permutation Tests, Pre-registered Design

*link:* https://github.com/drkianmaleki/reasoning-trace-experiments

*link:* https://github.com/drkianmaleki/decided-mid-thought

- *Reasoning Trace Experiments:* Designed a resampling instrument that completes a reasoning trace 25 times at each chosen block boundary, with an LLM judge labeling the completions.
- Registered the design before sampling, logged every departure, and re-derived headline numbers with three separate code paths.
- *Decided Mid-Thought:* Designed and pre-registered a causal study of whether a reasoning model's visible chain of thought reveals when its final answer becomes determined, using prefix interventions and 500 resampled continuations.
- Found no single-sentence commitment point against pre-registered thresholds; the default answer was $\sim$80% determined by the prompt alone, with 84.6% of the trace generated after the answer had saturated.

**LumberLex: Domain-Specific Product Normalization Engine** 2026

*Tech:* Python, RapidFuzz, Pydantic, Streamlit, pytest, Data Normalization, LLM Explanations

*link:* https://github.com/drkianmaleki/LumberLex

- Built a Python package and Streamlit application that normalizes inconsistent lumber product names into canonical structured outputs with confidence scores and manual-review flags, designed to solve a product-data normalization problem described by managers at a building-materials software company.
- Implemented fuzzy matching, dimension extraction, treatment detection, seller identification, and alternative matches, exposed through a Streamlit interface with chatbot-style explanations.

**End-to-End MLOps Pipeline: Heart Disease Classification** 2026

*Tech:* DVC, Git, MLflow, pytest, GitHub Actions, Evidently

*link:* https://github.com/drkianmaleki/Heart-Disease-Prediction

- Built a reproducible training-and-deployment workflow with data versioning (DVC), experiment tracking and a model registry (MLflow), and CI/CD validation gates.
- Automated testing with GitHub Actions and pytest, enforcing model-performance thresholds before deployment.
- Implemented data and concept drift monitoring with Evidently to support proactive retraining triggers.

**PDF Document Intelligence Extractor** 2026

*Tech:* Python, NLP, Regex, Pandas, PDF Text Extraction, RAG

*link:* https://github.com/drkianmaleki/PDF-Document-Intelligence-Extractor

- Built a document intelligence pipeline to extract key regulatory information from PDF documents using native text-layer parsing rather than OCR.
- Developed rule-based and NLP extraction methods to identify, normalize, and structure information from semi-structured text.

**AG News Classifier: End-to-End News Topic Classification** 2026

*Tech:* Python, scikit-learn, TF-IDF, MLflow, pytest, Docker, LLM API (optional)

*link:* https://github.com/drkianmaleki/AG-News-Classifier

- Developed an end-to-end classifier that assigns news articles to four topics, comparing five scikit-learn model configurations on TF-IDF features and selecting the best by weighted F1.
- Added MLflow experiment tracking, pytest tests for preprocessing, model, and interface behavior, and Docker packaging, with an optional LLM explanation layer kept separate from the classifier and a deterministic fallback when no API key is set.

## Selected Publications

- Maleki, K. et al. "Crystal fields, exchange and dipolar interactions..." *Physical Review B*, 2025.
- Kavand, M., Maleki, K., et al. "A General and Modular Approach to Solid-State Integration..." *Nano Letters*, 2025. Related to a patent-pending quantum device.